Monday Lab

Q1 Dothraki is planning an attack to usurp King Robert's throne. King Robert learns of this conspiracy from Raven and plans to lock the single door through which the enemy can enter his kingdom. But to lock the door, he needs a key that is an anagram of a palindrome. He starts to go through his box of strings, checking to see if they can be rearranged into a palindrome. You are given a string, write a function possiblePal(char str[ ]) which takes a string as an input and returns YES if the string can be rearranged into a palindrome else NO. The string will contain lowercase characters only. (Using dynamic memory allocation)

Example 1:
Input: aaabbbb
Output: YES
Explanation: The rearranged palindrome for the given string is bbaaabb
Example 2:
Input:
cdcdcdcdeeeef
Output:
YES
Explanation: The rearranged palindrome for the given string is ccddeefeeddcc

Q2. **Write a program that takes n students (name and roll number) as input using structures and then sorts the student list based on their roll numbers and display that.**

Test Data :
Input the string :
Enter number of student record to insert: 4
Enter name for student 1: ram
Enter roll number for student 1: 12
Enter name for student 2: shayam
Enter roll number for student 2: 5
Enter name for student 3: love
Enter roll number for student 3: 78
Enter name for student 4: kush
Enter roll number for student 4: 2
Expected Output :
Sorted list of students based on roll number :
Name: kush, Roll Number: 2
Name: shayam, Roll Number: 5
Name: ram, Roll Number: 12
Name: love, Roll Number: 78

Q3 Define a Vehicle structure with fields license_number, model, year, and owner. Inside owner, create a union to store either individual_name (if owned by a person) or company_name (if owned by a company). Write a function to input and display vehicle information, including the appropriate owner type.
Example
Displaying All Vehicles:

Vehicle Information:
License Number: ABC123
Model: Toyota Camry
Year: 2020
Owner (Individual): John Doe

Q4 Given an array arr[] containing N distances of inch-feet system, such that each element of the array represents a distance in the form of {inch, feet}. The task is to add all the N inch-feet distances using structures.
Example

Input: arr[] = { { 10, 3.7 }, { 10, 5.5 }, { 6, 8.0 } };
Output: Feet Sum: 27 Inch Sum: 5.20

Input: arr[] = { { 1, 1.7 }, { 1, 1.5 }, { 6, 8 } };
Output: Feet Sum: 8 Inch Sum: 11.20

Q5.Write a program to implement a student database using structures and perform operations like add, delete, search and display all student.
Example
Student Database Menu:
1. Add Student
2. Delete Student
3. Search Student
4. Display All Students
5. Exit
Enter your choice: 1

Enter roll number: 101
Enter name: Alice
Enter marks: 89.5
Student added successfully!

Student Database Menu:
1. Add Student
2. Delete Student
3. Search Student
4. Display All Students
5. Exit
Enter your choice: 4

Student Database:

Roll No	Name	Marks
101	Alice	89.50

Student Database Menu:
1. Add Student
2. Delete Student
3. Search Student
4. Display All Students
5. Exit
Enter your choice: 5
Exiting...